

1 Server & Database Overview

```
SELECT
@@SERVERNAME AS server_name,
sqlserver_start_time,
cpu_count,
physical_memory_kb / 1024 / 1024 AS memory_gb
FROM sys.dm_os_sys_info;
```

2 Current Blocking & Locks (CRITICAL)

```
SELECT
r.session_id,
r.blocking_session_id,
s.login_name,
s.host_name,
r.status,
r.wait_type,
r.wait_time / 1000 AS wait_seconds,
DB_NAME(r.database_id) AS database_name,
t.text AS blocked_query
FROM sys.dm_exec_requests r
JOIN sys.dm_exec_sessions s ON r.session_id = s.session_id
OUTER APPLY sys.dm_exec_sql_text(r.sql_handle) t
WHERE r.blocking_session_id <> 0
ORDER BY r.wait_time DESC;
```

3 Top Wait Types (Root Cause Analysis)

```
SELECT
wait_type,
waiting_tasks_count,
wait_time_ms / 1000 AS wait_seconds,
CAST(100.0 * wait_time_ms / SUM(wait_time_ms) OVER() AS DECIMAL(5,2)) AS
```

```
wait_percent
FROM sys.dm_os_wait_stats
WHERE wait_type NOT LIKE '%SLEEP%'
ORDER BY wait_time_ms DESC;
```

Key waits to watch

- LCK_M_* → Blocking
- PAGEIOLATCH_* → Disk IO
- CXPACKET → Parallelism
- SOS_SCHEDULER_YIELD → CPU pressure

4 Long-Running Queries

```
SELECT
r.session_id,
r.start_time,
r.status,
r.cpu_time,
r.total_elapsed_time / 1000 AS elapsed_seconds,
r.logical_reads,
r.writes,
DB_NAME(r.database_id) AS database_name,
t.text
FROM sys.dm_exec_requests r
OUTER APPLY sys.dm_exec_sql_text(r.sql_handle) t
ORDER BY r.total_elapsed_time DESC;
```

5 Most Expensive Queries (CPU & IO)

```
SELECT TOP 10
qs.execution_count,
qs.total_worker_time / qs.execution_count AS avg_cpu,
qs.total_elapsed_time / qs.execution_count AS avg_duration,
qs.total_logical_reads / qs.execution_count AS avg_reads,
DB_NAME(st.dbid) AS database_name,
st.text
FROM sys.dm_exec_query_stats qs
```

```
CROSS APPLY sys.dm_exec_sql_text(qs.sql_handle) st
ORDER BY avg_cpu DESC;
```

6 Table & Index Locks

```
SELECT
OBJECT_NAME(p.object_id) AS table_name,
tl.request_mode,
tl.request_status,
s.session_id,
s.login_name
FROM sys.dm_tran_locks tl
JOIN sys.dm_exec_sessions s ON tl.request_session_id = s.session_id
LEFT JOIN sys.partitions p ON tl.resource_associated_entity_id = p.hobt_id
WHERE tl.resource_type IN ('OBJECT','KEY','PAGE')
ORDER BY table_name;
```

7 Index Fragmentation (Performance Impact)

```
SELECT
OBJECT_NAME(ips.object_id) AS table_name,
i.name AS index_name,
ips.avg_fragmentation_in_percent,
ips.page_count
FROM sys.dm_db_index_physical_stats(DB_ID(), NULL, NULL, NULL, 'LIMITED') ips
JOIN sys.indexes i
ON ips.object_id = i.object_id AND ips.index_id = i.index_id
WHERE ips.page_count > 100
ORDER BY ips.avg_fragmentation_in_percent DESC;
```

Guideline

- 5–30% → Reorganize
- 30% → Rebuild

8 Disk IO Latency (Very Important)

```
SELECT
DB_NAME(vfs.database_id) AS database_name,
mf.name AS file_name,
vfs.num_of_reads,
vfs.io_stall_read_ms / NULLIF(vfs.num_of_reads, 0) AS avg_read_latency_ms,
vfs.num_of_writes,
vfs.io_stall_write_ms / NULLIF(vfs.num_of_writes, 0) AS avg_write_latency_ms
FROM sys.dm_io_virtual_file_stats(NULL, NULL) vfs
JOIN sys.database_files mf ON vfs.file_id = mf.file_id
ORDER BY avg_read_latency_ms DESC;
```

Warning

- 20 ms → Slow disk
- 50 ms → Critical

10 Database Size & Growth

```
SELECT
DB_NAME(database_id) AS database_name,
name AS file_name,
size * 8 / 1024 AS size_mb,
growth * 8 / 1024 AS growth_mb
FROM sys.database_files;
```

1 Top Slow Stored Procedures (By Duration)

```
SELECT TOP 10
DB_NAME(database_id) AS database_name,
OBJECT_NAME(object_id, database_id) AS procedure_name,
execution_count,
total_elapsed_time / execution_count / 1000 AS avg_duration_sec,
total_worker_time / execution_count / 1000 AS avg_cpu_sec,
total_logical_reads / execution_count AS avg_reads,
last_execution_time
```

```
FROM sys.dm_exec_procedure_stats  
ORDER BY avg_duration_sec DESC;
```

2 Stored Procedures Causing High CPU

```
SELECT TOP 10  
DB_NAME(database_id) AS database_name,  
OBJECT_NAME(object_id, database_id) AS procedure_name,  
execution_count,  
total_worker_time / execution_count / 1000 AS avg_cpu_sec,  
last_execution_time  
FROM sys.dm_exec_procedure_stats  
ORDER BY avg_cpu_sec DESC;
```

3 Stored Procedures Doing Heavy IO

```
SELECT TOP 10  
DB_NAME(database_id) AS database_name,  
OBJECT_NAME(object_id, database_id) AS procedure_name,  
execution_count,  
total_logical_reads / execution_count AS avg_reads,  
total_physical_reads / execution_count AS avg_physical_reads  
FROM sys.dm_exec_procedure_stats  
ORDER BY avg_reads DESC;
```

4 Currently Running Stored Procedures

```
SELECT  
r.session_id,  
r.status,  
r.start_time,  
r.cpu_time,  
r.total_elapsed_time / 1000 AS elapsed_sec,  
DB_NAME(r.database_id) AS database_name,  
t.text  
FROM sys.dm_exec_requests r
```

```
OUTER APPLY sys.dm_exec_sql_text(r.sql_handle) t
WHERE t.text LIKE '%PROC%'
ORDER BY r.total_elapsed_time DESC;
```

5 Stored Procedure Blocking

```
SELECT
r.session_id,
r.blocking_session_id,
DB_NAME(r.database_id) AS database_name,
t.text AS blocked_proc,
r.wait_type,
r.wait_time / 1000 AS wait_seconds
FROM sys.dm_exec_requests r
OUTER APPLY sys.dm_exec_sql_text(r.sql_handle) t
WHERE r.blocking_session_id <> 0;
```

6 Stored Procedures with Recompiles (Parameter Sniffing)

```
SELECT TOP 10
DB_NAME(database_id) AS database_name,
OBJECT_NAME(object_id, database_id) AS procedure_name,
cached_time,
last_execution_time,
execution_count,
total_elapsed_time / execution_count / 1000 AS avg_duration_sec
FROM sys.dm_exec_procedure_stats
WHERE execution_count > 1
ORDER BY cached_time DESC;
```

 Frequent recompiles may indicate:

- Parameter sniffing
- Table variable misuse
- Schema changes

7 Execution Plan for Slow Stored Procedure

```
SELECT
qs.execution_count,
qs.total_elapsed_time / qs.execution_count AS avg_duration,
qp.query_plan
FROM sys.dm_exec_query_stats qs
CROSS APPLY sys.dm_exec_query_plan(qs.plan_handle) qp
WHERE qs.sql_handle IN (
SELECT sql_handle
FROM sys.dm_exec_procedure_stats
);
```

8 Find Missing Indexes for Stored Procedures

```
SELECT
mid.statement AS table_name,
migs.avg_user_impact,
migs.user_seeks,
mid.equality_columns,
mid.inequality_columns,
mid.included_columns
FROM sys.dm_db_missing_index_group_stats migs
JOIN sys.dm_db_missing_index_groups mig
ON migs.group_handle = mig.index_group_handle
JOIN sys.dm_db_missing_index_details mid
ON mig.index_handle = mid.index_handle
ORDER BY migs.avg_user_impact DESC;
```